

# SASTRA DEEMED UNIVERSITY

(A University under section 3 of the UGC Act, 1956)

## End Semester Examinations

February 2022

Course Code: **CHY101**

Course: **ENGINEERING CHEMISTRY**

Question Paper No. : **UGF040**

Max. Marks: **100**

### PART - A

Answer all the questions

10 x 2 = 20 Marks

1. Water containing Fe & Mn ions cannot be treated using the Zeolite process. Why?
2. Hardness treatment by EDTA method uses basic buffer at pH~10. In presence of acidic buffer, no reaction is observed. Justify your understanding.
3. Calculate the EMF of Zn-Cu cell at 25°C with the following conditions.  $\text{Zn(s)}|\text{Zn}^{2+}(0.5\text{M})||\text{Cu}^{2+}(0.01\text{M})|\text{Cu(s)}$ ; ( $E^\circ=1.10\text{V}$ ).
4. Solid oxide and membrane fuel cells work in the presence of a  $\text{CO}_2$  environment, but alkaline fuel cells do not. Why?
5. Identify the molecules that possess dipole-dipole interactions in the following compounds:  
 $\text{CO}_2$ ,  $\text{CO}$ ,  $\text{ClF}$ ,  $\text{N}_2$
6. Write the significance of pour-point determination of oil.
7. How do you distinguish between particulate composite (coarse particles) and dispersion strengthened composite?

8. The particle sizes of 10 & 100 nm show different optical properties. Why?
9. Chloramine is a better choice for the sterilization of water than Chlorine. How?
10. Silver nitrate solution cannot be stored in a copper vessel. Justify.

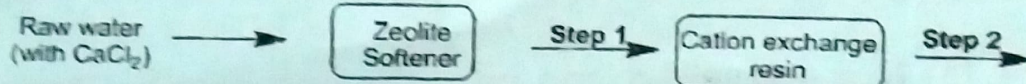
### **PART – B**

**Answer all the questions**

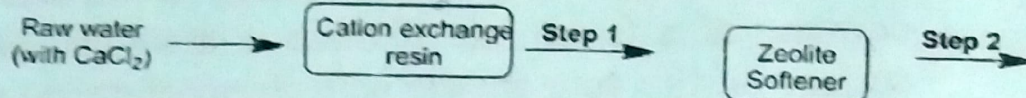
**4 x 15 = 60 Marks**

11. (a) SASTRA University collected waters for daily uses in and around Thanjavur and Trichy. Before the use of it, the university wanted to analyze water quality. As an engineering chemistry student, you have been assigned to check the water quality. For analysis, 1.53 g of  $\text{CaCO}_3$  was dissolved in HCl and the solution was made up to 1L with distilled water as a standard. 50 mL of the solution required 32 mL of EDTA solution for titration. 50 mL of hard water sample required 16 mL of EDTA and after boiling and filtering the same water required 9 mL of EDTA solution. Calculate all kinds of hardness in ppm. (7)
- (b) Assume that you are appointed as a technician for a water treatment plant in a reputed company. The required water was collected in different places. Based on the analysis report of waters' quality, they are divided into 4 schemes A to D as given below. Identify the purification process of each scheme in Step 1 and Step 2 with suitable reactions. Comment on the quality of water obtained after Step 2 in each scheme. Recommend the scheme which produces the best quality water. (8)

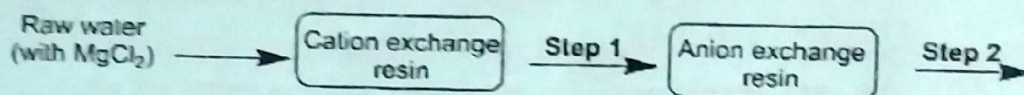
Scheme A



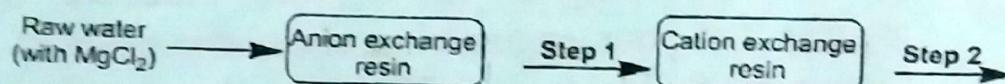
Scheme B



Scheme C



Scheme D



(OR)

12. (a) During summer days most of the metro cities in India are facing drinking water scarcity. To overcome the issues, Metro water departments have the plan to install the water treatment plants to treat seawater, so that the water demand will be somehow fulfilled. Suggest the best purification method for treating the seawater and strategy to remove biological impurities. Account the processes in detail. (7)

(b) A water sample on analysis gives the following data:  $\text{Ca}(\text{HCO}_3)_2 = 50$  ppm,  $\text{Mg}(\text{HCO}_3)_2 = 40$  ppm,  $\text{MgSO}_4 = 40$  ppm,  $\text{CaSO}_4 = 35$  ppm,  $\text{CaCl}_2 = 25$  ppm and  $\text{Na}_2\text{SO}_4 = 10$  ppm. The same water was treated with a zeolite softener and the softener was again regenerated when it was exhausted 80% using 100 L of NaCl solution of strength 150 g/L. Calculate the total hardness and volume of water treated. (8)

13. (a) For the galvanic cell:  $\text{Mg}_{(s)}|\text{Mg}^{2+}_{(aq)}(1\text{ M})||\text{Ag}^{+}_{(aq)}(x\text{ M})|\text{Ag}_{(s)}$ , the measured potential is 2.0V at 25°C. Calculate the concentration of  $[\text{Ag}^+]$   $\{E^\circ_{\text{Mg}/\text{Mg}^{2+}} = 2.37\text{V}; E^\circ_{\text{Ag}^+/\text{Ag}} = 0.80\text{V}\}$ . (7)

(b) Reliance Naval & Engineering Company Limited, Mumbai hired you as a chief mechanical engineer. If you have been allowed to construct the ship in a Shipyard, how do you protect the ship from corrosion by two different cathodic protection methods in the ocean? (8)

(OR)

14. (a) In a galvanized iron pipe, the iron does not form rust even the zinc surface is scratched. Whereas a tin-coated iron pipe corrodes at a faster rate if there is a scratch on a tin-coated surface. Justify your answer with electrochemical cells, reactions and respective calculations ( $E^\circ_{Fe/Fe^{2+}} = 0.44V$ ;  $E^\circ_{Zn/Zn^{2+}} = 0.76V$ ;  $E^\circ_{Sn/Sn^{2+}} = 0.14V$ ). (8)
- (b) As an engineering student at SASTRA University, you are currently preparing for the semester exam day and night. Just assume that your engineering chemistry exam is scheduled for tomorrow. Suddenly at 7 PM night, the electric power goes off due to some issues in EB line and the electricity board said that the problem will be resolved only by tomorrow morning. But your home is facilitated with battery backup. Elaborate on the working procedure of the method which helped you to overcome the power issue at home. (7)
15. (a) The molecules  $NH_3$ ,  $HF$ , and  $H_2O$  possess an equal number of electrons, but they show different boiling points  $-33.3$ ,  $19.5$ , and  $100^\circ C$ , respectively. Elaborate the reason for the different boiling points of these molecules. (7)
- (b) Our Indian Space Research Organization (ISRO) has planned to launch the space shuttle to the Mars in next few years. The local temperature of Mars is assumed to be very high. In this situation, what type of lubricant may be utilized for the machine parts which will work on Mars? Comment on the structure and function of the selected lubricant in detail. (8)

(OR)

16. (a) You have been given a set of oils with different boiling points. Assume that you are working in an extremely hot country and you have been asked to choose an appropriate lubricant that gives smooth machine functioning along with the local temperature. As you gained the knowledge of lubricants when you studied Engineering Chemistry, use your knowledge to identify a good lubricant by an appropriate experimental

procedure that will not catch the fire when the machine works. Justify your choice of lubricant based on the viscosity-index parameter. (6)

(b) (i) What is the viscosity index? Draw the mode of variation of viscosity with the temperature for low, test, and high viscosity oils. (8)

(ii) A lubricating oil shows the same Redwood viscosity as standard naphthenic and paraffinic oils at 210°F. Their Redwood viscosities at 100°F are 422 seconds, 590 seconds, and 415 seconds, respectively. Find the viscosity index of the given oil sample and comment on oil quality. (9)

17. (a) Qualitative analysis of drinking water confirms the presence of toxic elements As, Pb and microorganisms. Explain the working procedure of AgNPs based filter in removing the impurity & disease-causing microorganisms. (7)

(b) Distinguish the particle reinforcement and fiber reinforcement in composite materials. (8)

(OR)

18. (a) Elaborate on the structural reinforcement of composites in detail. (7)

(b) Explain the synthesis of nano-TiO<sub>2</sub> by sol-gel method and give the mechanism of self-cleaning action of nano-TiO<sub>2</sub> glass. (8)

### PART - C

**Answer the following**

**1 x 20 = 20 Marks**

19. (a) Presence of Na<sub>2</sub>CO<sub>3</sub>, MgCl<sub>2</sub>, and dissolved O<sub>2</sub> create huge problems in the high-pressure boilers. How do these cause boiler issues? Explain with respective chemical reactions. (6)

(b) Most of us are very fond of eating sweets. When you bite a sweet (aluminium foil coated) with a silver amalgam-filled tooth in the presence of saliva as a medium, you feel an electrical shock inside the mouth. Construct the electrochemical cell and write

the respective chemical reaction. Justify your observation with the help of the following parameters ( $E^\circ_{Ag^+/Ag} = 0.80V$ ;  $E^\circ_{Al/Al^{3+}} = 1.66V$ ).

(4)

(c) Rationalize the variation in physical properties for the following compounds based on the intermolecular interactions:

(5)

(i) Ethane has a boiling point of  $-89^\circ\text{C}$  while ethanol has a boiling point of  $78^\circ\text{C}$ .

(ii) Polythene cover is highly flexible while Teflon blocks are very hard materials.

(iii) Explain the synthesis of AuNPs by wet-reduction method and mechanism of curing of cancer by AuNPs based anticancer drug.

(5)

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